

Leo Suite: AI for Smarter Schools and Student Support

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Portfolio: leo-suite.vercel.app

GitHub: github.com/mentorkokkwa/leo-suite

Focus: AI engineering · product · school impact

Leo Suite explores real school problems — not toy demos. Three deployed applications cover **student learning** (EduLens), **campus robotics** (CampusBot), and **student wellbeing with safety guardrails** (YouthMentor).

Why This Portfolio Matters

- Real problems:** marking workload, robot path planning, youth mental-health AI risks
- Production engineering:** API routes, Zod validation, multi-provider LLM chains, mock fallbacks
- Responsible AI:** crisis guardrails, teacher-review flags, privacy-first local storage
- Live demos:** every project deployed — reviewers can click and test immediately

Three Projects

EduLens AI (Primary)

Structured homework feedback + syllabus-aware lesson generator.

- Marking-style JSON feedback & OCR
- Lesson pack: plan, worksheet, visuals
- PDF/DOCX/PPTX export

CampusBot AI

School service robot simulator with pathfinding.

- A* navigation + re-planning
- Delivery / guide / library tasks
- Metrics report

YouthMentor AI

Safety-first wellbeing mentor for students.

- 6-step structured reflection
- 3-layer crisis guardrails
- localStorage only — no cloud data

Live Demo Links (Click to Review)

Project	URL	Best entry point
Portfolio Home	leo-suite.vercel.app	All projects + PDFs
EduLens AI	leo-suite-edutech-six.vercel.app/edulens	Homework analyzer
CampusBot AI	leo-suite-robot.vercel.app/campusbot/simulator	Run simulation
YouthMentor AI	leo-suite-growth-swart.vercel.app/youthmentor/safety-demo	Safety walkthrough

Technical Highlights

Area	Evidence
Full-stack	Next.js 16, React 19, TypeScript, Tailwind CSS 4
AI engineering	Provider chain, RAG-lite syllabus, structured JSON + Zod
Algorithms	A* pathfinding, dynamic obstacle re-planning
Safety & ethics	Crisis-only UI; teacher-review flags; PDPA-minded design
Deployment	4 Vercel projects; mock mode for reliable school demos

School Contribution & Next Steps

Pilot EduLens for teacher lesson prep; explore CampusBot for campus logistics simulation; offer YouthMentor safety demo for counsellor review. Committed to responsible AI adoption with teachers always in the loop.